

Graphon Mean Field Games with Control Constraints Dependent upon State and Control Mean Fields

Ziqi Huang and Peter E. Caines

Abstract—In many real-world game-theoretic scenarios, individual agents face strict resource limitations. This work investigates resource-constrained interactions within the framework of Graphon Mean Field Games of Controls (GMFGCs), where agent dynamics and running costs depend on state and control mean fields. The local network joint state-control mean field serves as a macroscopic proxy for the local resource level and determines each agent’s moving feasible control set. The resulting system consists of coupled Hamilton–Jacobi–Bellman (HJB) and Fokker–Planck–Kolmogorov (FPK) equations. Under Hausdorff stability of the moving feasible sets, we establish the regularity of the minimized Hamiltonian and obtain a classical solution of the HJB equation. Uniform best-response sensitivity, together with a weak-coupling condition, makes the consistency mapping contractive, thereby establishing the existence and uniqueness of a consistent joint measure flow and, consequently, of a generalized Nash equilibrium. Numerical simulations with time-varying and fixed constraints illustrate optimal control saturation.

I. INTRODUCTION

Mean Field Games (MFGs) provide a framework for analyzing differential games involving a continuum of interacting agents. In realistic settings, agents frequently operate under strict state or control constraints. Incorporating such constraints introduces mathematical challenges. For instance, optimal controls induced by the cost structure may drive agents’ states toward the boundary of the feasible region, causing the state distribution to lose absolute continuity. The resulting singular measure cannot be handled by classical density-based partial differential equation (PDE) formulations. Consequently, one approach to constrained MFGs is to formulate Lagrangian equilibria as probability measures on state-control trajectories and study their existence, as in [1]–[4].

Existing literature has explored MFGs with constraints, particularly in deterministic settings. For example, crowd motion models restrict movement within specific spatial domains, utilizing accelerations as controls and incorporating congestion terms into the cost function [3]. To ensure well-posedness, regularity conditions on the initial state distributions and cost functions are typically proposed to guarantee that optimal trajectories do not encounter infeasible limiting directions. In such formulations, topological fixed-point theorems, such as Kakutani’s theorem, are employed to prove the existence of a Lagrangian equilibrium [4].

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Despite the success of direct Lagrangian formulations, the most straightforward approach remains solving the coupled Hamilton–Jacobi–Bellman (HJB) and Fokker–Planck–Kolmogorov (FPK) system to construct a contraction mapping. This methodology has been successfully extended to Graphon Mean Field Games (GMFGs) [5]–[7], which model heterogeneous, dense interactions over large-scale networks. In this work, we advance this paradigm by studying a Graphon Mean Field Game of Controls (GMFGC), in which the constraints on an agent’s control depend on the local network control mean field. This dynamic, distribution-dependent constraint fundamentally couples the feasible action space of an individual agent to the collective control behavior of its network neighbors, a feature that necessitates analyzing the joint probability measure of states and controls.

To tackle this coupled system, we impose Hausdorff stability of the moving feasible sets and a uniform stability condition on the complete best-response feedback. The latter accounts simultaneously for changes in the feasible set, Hamiltonian coefficients, and HJB costate; it is not inferred from Hausdorff stability alone. Under these assumptions we construct a fixed-point mapping on a complete space of path-measure ensembles. A weak-coupling condition makes this mapping contractive and uniquely determines the consistent joint measure flow within the stated class. Finally, we provide a numerical scheme and MATLAB simulations that illustrate the theoretical findings, highlighting the distinct macroscopic behaviors induced by mean-field-dependent control constraints.

II. PRELIMINARIES

A. Notation

This paper studies the following HJB–FPK system for a GMFG with control constraints depending on the local network state–control mean field.

$$\begin{aligned} [HJB](\alpha) \quad & -\frac{\partial V^\alpha(t, x)}{\partial t} = \inf_{u \in U_\alpha(t, \bar{\lambda}_\alpha)} \left\{ \tilde{l}(x, u, \nu_G; g_\alpha) \right. \\ & \left. + \tilde{f}(x, u, \nu_G; g_\alpha) \cdot \nabla V^\alpha(t, x) \right\} + \frac{\sigma^2}{2} \Delta V^\alpha(t, x), \\ V^\alpha(T, x) &= g(x), \\ [FPK](\alpha) \quad & \frac{\partial p_\alpha(t, x)}{\partial t} = -\nabla \cdot \left(\tilde{f}(x, u^\circ, \nu_G; g_\alpha) p_\alpha(t, x) \right) \\ & + \frac{\sigma^2}{2} \Delta p_\alpha(t, x), \quad p_\alpha(0, \cdot) = \mu_0 \end{aligned} \quad (1)$$

$$[BR](\alpha) \quad u^\circ := \varphi(t, x, \nu_G; g_\alpha), \quad (2)$$

where $\nu_G \in C([0, 1], \mathcal{P}(\mathbb{R}^n \times \mathbb{R}^m))$ is the joint probability measure ensemble of state and control at a specific time, λ_α is the network state and control mean field that will be defined in (8), $U_\alpha(t, \lambda_\alpha)$ is the time wise feasible control set that will be defined in (6), and the functions $\tilde{f}$ and $\tilde{l}$ will be defined later.

Let $\alpha \in [0, 1]$ be a node index in a graphon. Let $n, m \in \mathbb{N}$ denote the state and control dimensions, respectively; in general, m need not equal n . Let $[0, T]$ be the time horizon. The value of the a state path, a control path and a Brownian motion at node α at time t are denoted as $x_\alpha(t) \in \mathbb{R}^n$, $u_\alpha(t) \in \mathbb{R}^m$, and $W_\alpha(t) \in \mathbb{R}^n$ respectively.

Let $\mu_{\alpha,t} \in \mathcal{P}_1(\mathbb{R}^n)$ be the probability measure of the state variable at node α and time t , where $\mathcal{P}_1(\mathbb{R}^n)$ is a probability measure space with the 1-Wasserstein distance. Define the joint probability measure $\nu_{\alpha,t} \in \mathcal{P}_1(\mathbb{R}^n \times \mathbb{R}^m)$, which is the joint probability measure of the state and control.

Let $\mu_\alpha := \{\mu_{\alpha,t}\}_{0 \leq t \leq T} \in C([0, T], \mathcal{P}_1(\mathbb{R}^n))$ and $\nu_\alpha := \{\nu_{\alpha,t}\}_{0 \leq t \leq T} \in C([0, T], \mathcal{P}_1(\mathbb{R}^n \times \mathbb{R}^m))$ be the continuous probability measure flow of the state at node α and continuous joint probability measure flow of state and control at node α respectively.

Let $\mu_G := \{\mu_\alpha\}_{\alpha \in [0,1]}$ and $\nu_G := \{\nu_\alpha\}_{\alpha \in [0,1]}$ be the probability measure flow ensemble of the state and joint probability measure flow ensemble of state and control respectively. Let $\mu_G(t)$ and $\nu_G(t)$ be the probability measure ensemble of the state variable at time t , and the joint probability measure ensemble of state and control at time t respectively. We assume that $\mu_G(t) \in C([0, 1], \mathcal{P}_1(\mathbb{R}^n))$ and $\nu_G(t) \in C([0, 1], \mathcal{P}_1(\mathbb{R}^n \times \mathbb{R}^m))$ for all $t \in [0, T]$.

Let $(\Omega, \mathcal{F}, \mathbb{F}_\alpha := \{\mathcal{F}_{\alpha,t}\}_{0 \leq t \leq T}, \mathbb{P}_\alpha)$ be the probability space at node α . And the Brownian motion and state trajectory at node α are adapted to $\mathbb{F}_\alpha$. Assume that the Brownian motions $w_\alpha(t)$ are independent for different α .

B. Problem Setting

Suppose that each node α contains an infinite population. A generic agent at node α faces the stochastic control problem

$$\inf_{\mathbf{u}_\alpha \in \mathbb{A}_\alpha(\nu_G)} J_\alpha^{\nu_G}(\mathbf{x}_\alpha, \mathbf{u}_\alpha) := \mathbb{E} \left[\int_0^T \tilde{l}(x_\alpha^{\mathbf{u}_\alpha}(t), u_\alpha(t), \nu_G(t); g_\alpha) dt + g(x_\alpha^{\mathbf{u}_\alpha}(T)) \right]$$

subject to

$$\begin{aligned} dx_\alpha^{\mathbf{u}_\alpha}(t) &= \tilde{f}(x_\alpha^{\mathbf{u}_\alpha}(t), u_\alpha(t), \nu_G(t); g_\alpha) dt + \sigma dw_\alpha(t), \\ x_\alpha^{\mathbf{u}_\alpha}(0) &= \xi. \end{aligned} \quad (3)$$

where $\mathbf{x}_\alpha, \mathbf{u}_\alpha$ denote functions on the time horizon $[0, T]$, and l, g, f, σ are functions of appropriate dimensions, and the feasible control space $\mathbb{A}_\alpha$ depends on the joint measure ensemble ν_G and is defined as follows,

$$\begin{aligned} \mathbb{A}_\alpha(\nu_G) &:= \left\{ \mathbf{u}_\alpha : \Omega \times [0, T] \rightarrow \mathbb{R}^m \mid \mathbb{E} \int_0^T |u_\alpha(t)|^2 dt < \infty, \right. \\ &\quad \mathbb{F}_\alpha\text{-progressively measurable,} \\ &\quad \left. c_\alpha(\mathbf{u}_\alpha, \nu_G) \leq 0, \text{ dt} \otimes d\mathbb{P}\text{-a.e.} \right\} \end{aligned} \quad (4)$$

where $c_\alpha : L^2(\Omega \times [0, T]; \mathbb{R}^m) \times \prod_{\alpha \in [0,1]} C([0, T], \mathcal{P}_1(\mathbb{R}^n \times \mathbb{R}^m)) \ni (\mathbf{u}_\alpha, \nu_G) \mapsto c_\alpha(\mathbf{u}_\alpha, \nu_G) \in L^\infty(\Omega \times [0, T], \mathbb{R}^{n_c})$ and Leb_1 is the Lebesgue measure on the real line. Let $\mathbf{x}_G := \{\mathbf{x}_\alpha\}_{\alpha \in [0,1]}$ and $\mathbf{u}_G := \{\mathbf{u}_\alpha\}_{\alpha \in [0,1]}$ be the trajectory ensemble of state and control respectively.

Restrict the pathwise constraint c_α to the pointwise-in-time constraint c_α defined as $\mathbb{R}^m \times C([0, 1], \mathcal{P}_1(\mathbb{R}^n \times \mathbb{R}^m)) \ni (u_\alpha(t), \nu_G(t)) \mapsto c_\alpha(u_\alpha(t), \nu_G(t)) \in \mathbb{R}^{n_c}$. The Hamiltonian of the system at node α and time t would take the following form,

$$\begin{aligned} H_\alpha(t, x_\alpha(t), u_\alpha(t), \nu_G(t), p_\alpha(t)) &:= \tilde{l}(x_\alpha(t), u_\alpha(t), \nu_G(t)) \\ &\quad + p_\alpha(t) \cdot \tilde{f}(x_\alpha(t), u_\alpha(t), \nu_G(t)) \end{aligned} \quad (5)$$

where $p_\alpha(t) \in \mathbb{R}^n$.

Define the following time-wise feasible set of control,

$$A_\alpha(t, \nu_G(t)) := \left\{ u_\alpha(t) \in \mathbb{R}^m \mid c_\alpha(u_\alpha(t), \nu_G(t)) \leq 0 \right\} \quad (6)$$

Define the graphon integral operation on the measure spaces. For any trajectory ensembles $\mathbf{x}_G \in L^2([0, 1] \times [0, T]; \mathbb{R}^n)$ and $\mathbf{u}_G \in L^2([0, 1] \times [0, T]; \mathbb{R}^m)$, any measure ensemble μ_G, ν_G , and any graphon $\mathcal{W} \in L^2([0, 1]^2)$, we define the following network mean fields,

$$\bar{\kappa}_\alpha(t) := \int_0^1 \mathcal{W}(\alpha, \beta) \int_{\mathbb{R}^n} x_\beta(t) \mu_{\beta,t}(dx_\beta(t)) d\beta, \quad (7)$$

$$\begin{aligned} \bar{\lambda}_\alpha(t) &:= \int_0^1 \mathcal{W}(\alpha, \beta) \int_{\mathbb{R}^n \times \mathbb{R}^m} (x_\beta(t), u_\beta(t)) \\ &\quad \nu_{\beta,t}(dx_\beta(t), du_\beta(t)) d\beta, \end{aligned} \quad (8)$$

so the above definitions would give us $\bar{\kappa}_\alpha \in L^2([0, T]; \mathbb{R}^n)$, $\bar{\lambda}_\alpha \in L^2([0, T]; \mathbb{R}^{n+m})$, $\bar{\kappa}_t \in L^2([0, 1]; \mathbb{R}^n)$, $\bar{\lambda}_t \in L^2([0, 1]; \mathbb{R}^{n+m})$, $\bar{\kappa}_G \in L^2([0, 1] \times [0, T]; \mathbb{R}^n)$ and $\bar{\lambda}_G \in L^2([0, 1] \times [0, T]; \mathbb{R}^{n+m})$. Higher moments may also be considered; this paper uses only first moments.

Suppose that the constraints c_α only depend on the measure ensemble $\nu_G(t)$ through $\bar{\lambda}_\alpha(t)$, and define the following equivalent constraints depending on $\bar{\lambda}_\alpha(t)$,

$$\gamma_\alpha(u_\alpha(t), \bar{\lambda}_\alpha(t)) := c_\alpha(u_\alpha(t), \nu_G(t)). \quad (9)$$

We have the definition of the feasible control set

$$U_\alpha(t, \bar{\lambda}_\alpha(t)) := \left\{ u_\alpha(t) \in \mathbb{R}^m \mid \gamma_\alpha(u_\alpha(t), \bar{\lambda}_\alpha(t)) \leq 0 \right\}. \quad (10)$$

We follow a similar setting with [5] by splitting $\tilde{f}, \tilde{l}$ into a local part and a graphon integration part. Write $g_\alpha(\beta) := \mathcal{W}(\alpha, \beta)$ for the graphon row at node α .

Let f and $\tilde{l}$ be defined as follows,

$$\begin{aligned} \tilde{f}(x, u, \nu_G; g_\alpha) &:= \int_{\mathbb{R}^n \times \mathbb{R}^m} f_0(x, u, y, v) d\nu_\alpha(y, v) \\ &\quad + \int_0^1 \mathcal{W}(\alpha, \beta) \int_{\mathbb{R}^n \times \mathbb{R}^m} f(x, u, y, v) \nu_\beta(dy, dv) d\beta, \end{aligned} \quad (11)$$

$$\begin{aligned} \tilde{l}(x, u, \nu_G; g_\alpha) &:= \int_{\mathbb{R}^n \times \mathbb{R}^m} l_0(x, u, y, v) d\nu_\alpha(y, v) \\ &+ \int_0^1 \mathcal{W}(\alpha, \beta) \int_{\mathbb{R}^n \times \mathbb{R}^m} l(x, u, y, v) \nu_\beta(dy, dv) d\beta, \end{aligned} \quad (12)$$

where the assumptions on the functions f_0, l_0, f, l will be specified below.

C. Assumptions

The letter C denotes a positive constant that may change from line to line.

(H1) For each admissible $\nu_G \in \mathcal{V}_{[0,T]}$, let $a_\alpha(t, x, p | \nu_G)$ be the unique minimizer of $\tilde{f}(x, u, \nu_G(t); g_\alpha) \cdot p + \tilde{l}(x, u, \nu_G(t); g_\alpha)$ over $u \in U_\alpha(t, \bar{\lambda}_\alpha^\nu(t))$. For every $P > 0$ there is $L_a(P)$, uniform in α, t, ν_G , such that $|a_\alpha(t, x, p | \nu_G) - a_\alpha(t, y, q | \nu_G)| \leq L_a(P)(|x - y| + |p - q|)$ for $|p|, |q| \leq P$.

(H2) There is a compact convex $K \subset \mathbb{R}^m$ such that $\emptyset \neq U_\alpha(t, \lambda) \subset K$ for every admissible (α, t, λ) .

(H3) For some $L_U > 0$ and modulus ω_U ,

$$d_H(U_\alpha(t, \lambda), U_\beta(s, \eta)) \leq L_U(|t - s| + |\lambda - \eta|) + \omega_U(|\alpha - \beta|). \quad (13)$$

Uniform LICQ and second-order sufficient conditions imply (H1)–(H3) for the constraint-qualified models considered in [8].

(H4) The functions $h \in \{f_0, f, l_0, l\}$ are bounded and globally Lipschitz: $|h(x, u, y, v)| \leq M_h$ and $|h(z) - h(z')| \leq L_h|z - z'|$ for $z = (x, u, y, v)$, with constants uniform in α .

(H5) For some $\theta \in (0, 1)$, $g \in C_b^{2+\theta}(\mathbb{R}^n)$ and $\sigma > 0$. All nodes have the common initial law $\mu_0 \in \mathcal{P}_1(\mathbb{R}^n)$; when (1) is written in density form, μ_0 also denotes its density.

(H6) For every bounded measurable $\zeta : [0, 1] \rightarrow \mathbb{R}^d$, $d \in \{1, n, n + m\}$, the map $\alpha \mapsto \int_0^1 \mathcal{W}(\alpha, \beta) \zeta(\beta) d\beta$ is continuous.

(H7) The kernel satisfies $M_{\mathcal{W}} := \sup_{\alpha \in [0, 1]} \int_0^1 |\mathcal{W}(\alpha, \beta)| d\beta < \infty$.

Admissible flows. Fix once and for all $\eta \in (0, 1/2]$ and constants $C_h, M_V > 0$. Let $\mathcal{V}_{[0,T]}$ contain the joint flows $\nu_G = (\nu_{\alpha,t})$ satisfying: (C1) $\alpha \mapsto \nu_{\alpha,t}$ is continuous in W_1 for every t ; (C2) $\sup_\alpha W_1(\nu_{\alpha,t}, \nu_{\alpha,s}) \leq C_h|t - s|^\eta$; and (C3) $\sup_{\alpha,t} \int |(x, u)| \nu_{\alpha,t}(dx, du) \leq M_V$. Equip this class with

$$d_{\mathcal{V}}(\nu_G, \nu'_G) := \sup_{\alpha,t} W_1(\nu_{\alpha,t}, \nu'_{\alpha,t}). \quad (14)$$

The constants above are fixed class parameters, so (C1)–(C3) are closed under $d_{\mathcal{V}}$ convergence.

Path laws. Set $C_T = C([0, T], \mathbb{R}^n \times \mathbb{R}^m)$ with $\rho(\omega, \omega') = \sup_{t \leq T} |\omega(t) - \omega'(t)|$ and its Borel σ -field. Let $\mathbf{V}_{\alpha,T} = \mathcal{P}_1(\bar{C}_T)$ and, for $t \in [0, T]$,

$$D_t(\nu_1, \nu_2) := \inf_{\hat{\pi}} \int_{C_T^2} \sup_{s \leq t} |X_s(\omega_1) - X_s(\omega_2)| d\hat{\pi}. \quad (15)$$

Here $\hat{\pi}$ ranges over couplings. Then D_T is the 1-Wasserstein metric induced by ρ ; here $X_s(\omega) := \omega(s) \in \mathbb{R}^{n+m}$ is the joint canonical coordinate. Moreover, $(\mathbf{V}_{\alpha,T}, D_T)$ is complete [9]. Define $\mathbf{V}_T^G = C([0, 1], \mathcal{P}_1(C_T))$ with the uniform metric $D_T(\nu_G, \tilde{\nu}_G) = \sup_\alpha D_T(\nu_\alpha, \tilde{\nu}_\alpha)$.

Lemma 1: The space $(\mathbf{V}_T^G, D_T)$ is complete.

Proof: A uniformly Cauchy sequence has a pointwise limit in the complete space $\mathcal{P}_1(C_T)$. Uniform convergence preserves continuity in α and gives convergence in the supremum metric. ■

III. ANALYSIS ON THE HJB EQUATION WITH A GIVEN PROBABILITY MEASURE FLOW

We will construct a contraction mapping on the above complete metric space $\mathbf{V}_T^G$. For the HJB analysis, we start from a fixed joint probability flow ensemble $\nu_G(\cdot) \in \mathcal{V}_{[0,T]}$, which yields a classical solution to the HJB, as shown in Lemma 4.

Note that we can always select such a probability measure flow ensemble despite there are constraints on the control, which would make its distribution at some time instant not absolutely continuous; nevertheless, the probability measure always exists. We use the following notation for the fixed dynamics, running costs, and network control mean fields,

$$\begin{aligned} \tilde{f}_\alpha^*(t, x, u) &:= \tilde{f}(x, u, \nu_G; g_\alpha), \quad \tilde{l}_\alpha^*(t, x, u) := \tilde{l}(x, u, \nu_G; g_\alpha), \\ \bar{\lambda}_\alpha^*(t) &:= \int_0^1 \mathcal{W}(\alpha, \beta) \int_{\mathbb{R}^n \times \mathbb{R}^m} (x_\beta(t), u_\beta(t)) \\ &\quad \nu_{\beta,t}(dx_\beta(t), du_\beta(t)) d\beta. \end{aligned} \quad (16)$$

For this fixed $\nu_G(\cdot)$, define the minimized Hamiltonian by

$$\hat{H}_\alpha(t, x, p) := \min_{u \in U_\alpha(t, \bar{\lambda}_\alpha^*)} \tilde{f}_\alpha^*(t, x, u) \cdot p + \tilde{l}_\alpha^*(t, x, u). \quad (17)$$

The HJB equation is therefore

$$\begin{aligned} -V_t^\alpha(t, x) &= \hat{H}_\alpha(t, x, \nabla V^\alpha(t, x)) + \frac{\sigma^2}{2} \Delta V^\alpha(t, x), \\ V^\alpha(T, x) &= g(x). \end{aligned} \quad (18)$$

We have the following Lipschitz and Hölder continuity properties on the dynamics $\tilde{f}_\alpha^*(t, x, u)$ and cost function $\tilde{l}_\alpha^*(t, x, u)$.

Lemma 2: For $h_\alpha = \tilde{f}_\alpha^*(t, x, u)$ or $\tilde{l}_\alpha^*(t, x, u)$, there exist positive constants $C_x, C_u, C_t > 0$ such that h_α is Lipschitz continuous in x and u with Lipschitz constants C_x and C_u , and Hölder continuous in t with constant C_t and exponent η .

Proof: We demonstrate the proof of $\tilde{f}_\alpha^*$, and the proof for $\tilde{l}_\alpha^*$ is identical.

(i) Lipschitz continuity in x and u :

By assumption (H4), the base functions f_0 and f are Lipschitz continuous in (x, u) uniformly with respect to (y, v) . Let L_f be the maximum of their Lipschitz constants. For any $x_1, x_2 \in \mathbb{R}^n$, we have,

$$\begin{aligned} &|\tilde{f}_\alpha^*(t, x_1, u) - \tilde{f}_\alpha^*(t, x_2, u)| \\ &\leq \int_{\mathbb{R}^{n+m}} |f_0(x_1, \cdot) - f_0(x_2, \cdot)| d\nu_{\alpha,t} \\ &\quad + \int_0^1 |\mathcal{W}(\alpha, \beta)| \int_{\mathbb{R}^{n+m}} |f(x_1, \cdot) - f(x_2, \cdot)| d\nu_{\beta,t} d\beta \\ &\leq L_f(1 + M_{\mathcal{W}})|x_1 - x_2|. \end{aligned} \quad (19)$$

Since the probability measures integrate to 1, and by (H7) the graphon integral is bounded by some constants $M_{\mathcal{W}}$, we obtain $|\tilde{f}_\alpha^*(t, x_1, u) - \tilde{f}_\alpha^*(t, x_2, u)| \leq L_f(1 + M_{\mathcal{W}})|x_1 - x_2|$. The exact same argument applies to u .

(ii) Hölder continuity in t :

By the definition of the fixed environment space (C2), the probability measure flow $t \mapsto \nu_G(t)$ is Hölder continuous under Wasserstein metric W_1 with exponent η . Since f_0 and f are Lipschitz continuous in (y, v) , we can use the Kantorovich- Rubinstein duality for the Wasserstein distance. For any $t, s \in [0, T]$,

$$|\tilde{f}_\alpha^*(t, x, u) - \tilde{f}_\alpha^*(s, x, u)| \leq \text{Lip}(f_0)W_1(\nu_{\alpha,t}, \nu_{\alpha,s}) + M_{\mathcal{W}} \sup_{\beta} (\text{Lip}(f)W_1(\nu_{\beta,t}, \nu_{\beta,s})). \quad (20)$$

By the Hölder property of the measure flow, $W_1(\nu_{\beta,t}, \nu_{\beta,s}) \leq C_h|t - s|^\eta$. Therefore,

$$|\tilde{f}_\alpha^*(t, x, u) - \tilde{f}_\alpha^*(s, x, u)| \leq C|t - s|^\eta, \quad (21)$$

completing the proof. $\blacksquare$

We need some regularity properties of the minimized Hamiltonian $\hat{H}_\alpha(t, x, p)$ in order to apply the linear PDE theory from [10].

Lemma 3: For any $\alpha \in [0, 1]$, any x, y, p, q in a compact subset of $\mathbb{R}^n$, and any $t, s \in [0, T]$, there exist constants $C, \eta > 0$ such that the following inequalities are satisfied,

$$\begin{aligned} |\hat{H}_\alpha(t, x, p) - \hat{H}_\alpha(s, x, p)| &\leq C(1 + |p|)|t - s|^\eta, \\ |\hat{H}_\alpha(t, x, p) - \hat{H}_\alpha(t, y, p)| &\leq C(1 + |p|)|x - y|, \\ |\hat{H}_\alpha(t, x, p) - \hat{H}_\alpha(t, x, q)| &\leq C|p - q|. \end{aligned} \quad (22)$$

Moreover, $\hat{H}_\alpha(t, x, p)$ is continuous in α .

Proof: Let the pre-minimized function be denoted as $J_\alpha := \tilde{f}_\alpha^*(t, x, u) \cdot p + \tilde{l}_\alpha^*(t, x, u)$. By Lemma 2, J_α is Lipschitz in x, u, p and Hölder continuous in t . By (H2), the control u is constrained in a uniformly compact set K .

(i) Lipschitz continuity in p :

$$\begin{aligned} &|\hat{H}_\alpha(t, x, p) - \hat{H}_\alpha(t, x, q)| \\ &= \left| \min_{u \in U_\alpha} J_\alpha(t, x, u, p) - \min_{u \in U_\alpha} J_\alpha(t, x, u, q) \right| \\ &\leq \max_{u \in U_\alpha} |J_\alpha(t, x, u, p) - J_\alpha(t, x, u, q)| \\ &\leq \max_{u \in K} |\tilde{f}_\alpha^*(t, x, u) \cdot (p - q)| \\ &\leq \|\tilde{f}_\alpha^*\|_\infty |p - q| \leq C|p - q|, \end{aligned} \quad (23)$$

where the boundedness of $\tilde{f}_\alpha^*$ follows from (H4).

(ii) Lipschitz continuity in x :

Let u_x and u_y be the minimizer of $\tilde{H}_\alpha$ evaluated at x and y , respectively. Then,

$$\begin{aligned} \hat{H}_\alpha(t, x, p) - \hat{H}_\alpha(t, y, p) &\leq J_\alpha(t, x, u_x, p) - J_\alpha(t, y, u_y, p) \\ &= (\tilde{f}_\alpha^*(t, x, u_x) - \tilde{f}_\alpha^*(t, y, u_y)) \cdot p \\ &\quad + (\tilde{l}_\alpha^*(t, x, u_x) - \tilde{l}_\alpha^*(t, y, u_y)). \end{aligned} \quad (24)$$

By the Lipschitz continuity of $\tilde{f}_\alpha^*$ and $\tilde{l}_\alpha^*$ in x established in Lemma 2 (with constant C_x),

$$\begin{aligned} \hat{H}_\alpha(t, x, p) - \hat{H}_\alpha(t, y, p) &\leq C_x|x - y||p| + C_x|x - y| \\ &\leq C_x(1 + |p|)|x - y|. \end{aligned} \quad (25)$$

Reversing the roles of x and y yields the symmetric lower bound, establishing $|\hat{H}_\alpha(t, x, p) - \hat{H}_\alpha(t, y, p)| \leq C(1 + |p|)|x - y|$.

(iii) Hölder continuity in t :

This step is needed for moving constraints. Let $u_s \in U_\alpha(s, \bar{\lambda}_\alpha(s))$ be the minimizer at time s . By (H3), the constraint set is Lipschitz in the Hausdorff metric d_H with respect to $(t, \bar{\lambda}_\alpha)$. Since $\nu_G(\cdot)$ is Hölder continuous in time, $\bar{\lambda}_\alpha(t)$ is also η -Hölder in time. Thus,

$$\begin{aligned} d_H(U_\alpha(t, \bar{\lambda}_\alpha(t)), U_\alpha(s, \bar{\lambda}_\alpha(s))) \\ \leq C(|t - s| + |\bar{\lambda}_\alpha(t) - \bar{\lambda}_\alpha(s)|) \leq C_1|t - s|^\eta. \end{aligned} \quad (26)$$

This implies that there exists a feasible control at time t , $u_t \in U_\alpha(t, \bar{\lambda}_\alpha(t))$, such that $|u_t - u_s| \leq C_1|t - s|^\eta$.

We bound the Hamiltonian difference by evaluating the t -Hamiltonian at this suboptimal u_t ,

$$\hat{H}_\alpha(t, x, p) - \hat{H}_\alpha(s, x, p) \leq J_\alpha(t, x, u_t, p) - J_\alpha(s, x, u_s, p).$$

Add and subtract $J_\alpha(s, x, u_t, p)$ in the preceding expression,

$$\begin{aligned} &[J_\alpha(t, x, u_t, p) - J_\alpha(s, x, u_t, p)] \\ &+ [J_\alpha(s, x, u_t, p) - J_\alpha(s, x, u_s, p)]. \end{aligned} \quad (27)$$

The first bracket is bounded by $C_2(1 + |p|)|t - s|^\eta$ due to the Hölder continuity of $\tilde{f}_\alpha^*$ and $\tilde{l}_\alpha^*$ in time (Lemma 2). The second bracket is bounded by $C_u(1 + |p|)|u_t - u_s|$ due to the Lipschitz continuity in u (Lemma 2).

Substituting the Hausdorff bound $|u_t - u_s| \leq C_1|t - s|^\eta$,

$$\begin{aligned} \hat{H}_\alpha(t, x, p) - \hat{H}_\alpha(s, x, p) &\leq C_2(1 + |p|)|t - s|^\eta \\ &\quad + C_u C_1(1 + |p|)|t - s|^\eta \leq C(1 + |p|)|t - s|^\eta. \end{aligned} \quad (28)$$

Symmetry yields the absolute value.

(iv) Continuity in α :

To establish continuity in α , we note that the inner integral of the network control mean field $\zeta(\beta) := \int u d\nu_{\beta,t}$ is a bounded measurable function due to the compactness of the control space (H2). By the property of the graphon assumed in (H6), the mapping $\alpha \mapsto \bar{\lambda}_\alpha(t) = \int_0^1 \mathcal{W}(\alpha, \beta) \zeta(\beta) d\beta$ is continuous. Since the constraint set U_α is continuous with respect to $\bar{\lambda}_\alpha$ in the Hausdorff metric (H3), and the basis functions are uniformly continuous, the minimized Hamiltonian $\hat{H}_\alpha$ inherits continuity in α . $\blacksquare$

The established regularity yields the following standard HJB result [5].

Lemma 4 (Solvability of HJB): Let assumptions (H1)–(H7) hold. For any fixed joint measure flow $\nu_G \in \mathcal{V}_{[0,T]}$, the resulting terminal value problem for the HJB equation (18) admits a unique bounded classical solution $V^\alpha \in C_b([0, T] \times \mathbb{R}^n) \cap C_{\text{loc}}^{1+\rho/2, 2+\rho}([0, T] \times \mathbb{R}^n)$. Here ρ is any number in $(0, \min\{\theta, 2\eta\})$. Furthermore, there exists a constant C_V , independent of α and ν_G , such that the spatial gradient is

uniformly bounded and globally Lipschitz in x ,

$$\sup_{\alpha, t, x} |\nabla V^\alpha(t, x)| \leq C_V,$$

$$\sup_{\alpha, t, x \neq y} \frac{|\nabla V^\alpha(t, x) - \nabla V^\alpha(t, y)|}{|x - y|} \leq C_V.$$

Proof: Fix α and ν_G , put $a = \sigma^2/2$, and write $H(t, x, p) = \hat{H}_\alpha(t, x, p)$. Lemma 3 gives $|H(t, x, 0)| \leq C$, a global Lipschitz bound in p , and local parabolic ρ -Hölder continuity in (t, x) on bounded p -sets. After the change $r = T - t$, $W(r, x) = V^\alpha(T - r, x)$ solves $W_r - a\Delta W = H(T - r, x, DW)$, $W(0) = g$. For the heat semigroup (P_r) define

$$(\mathcal{T}w)(r) = P_r g + \int_0^r P_{r-s} H(T - s, \cdot, Dw(s)) ds.$$

On the space of bounded functions with bounded uniformly continuous gradient, $\|P_r h\|_\infty \leq \|h\|_\infty$ and $\|DP_r h\|_\infty \leq C r^{-1/2} \|h\|_\infty$ give $\|\mathcal{T}w - \mathcal{T}z\| \leq C(\tau + \sqrt{\tau}) \|w - z\|$ on $[0, \tau]$. Thus $\mathcal{T}$ is a contraction for a uniform $\tau > 0$ and may be restarted finitely many times to cover $[0, T]$.

Heat-kernel difference estimates first yield local Hölder continuity of DW ; the whole-space Cauchy Schauder theorem [10, Ch. IV, Sec. 5, Thm. 5.1] then gives $W \in C_{\text{loc}}^{1+\rho/2, 2+\rho}$. The maximum principle [10, Ch. I, Sec. 2, Thm. 2.5] gives a uniform L^∞ bound. Applying Itô's formula up to exit times from balls shows that every admissible control has cost at least V^α ; a measurable Hamiltonian minimizer attains equality. Writing the bounded drift change as $\sigma\Theta$, the absolutely continuous change of measure in [11, Ch. V, Sec. 10, Thm. 10.2]—the Girsanov construction—gives weak existence for the closed-loop SDE. The unique minimizer in (H1) is continuous in (x, p) on the relevant bounded p -range and measurable in t ; compactness of K therefore makes the feedback progressively measurable and square integrable. Localization at $\tau_R = \inf\{s : |X_s| \geq R\} \wedge T$, followed by dominated convergence as $R \rightarrow \infty$, removes the stopping because the coefficients and feedback are bounded. Consequently V^α is the value function. For initial states x, y , applying the same admissible control and Brownian path gives $\mathbb{E} \sup_{s \leq T} |X_s^x - X_s^y| \leq C|x - y|$; the Lipschitz running and terminal data then imply $|V^\alpha(t, x) - V^\alpha(t, y)| \leq C|x - y|$, uniformly in α and the prescribed flow. This proves the stated bound on DV^α .

Finally, on this bounded gradient range, $\text{Lip}_x H(T - r, \cdot, DW) \leq C(1 + \|D^2 W(r)\|_\infty)$. Using $\|D^2 P_r h\|_\infty \leq C r^{-1/2} \text{Lip}(h)$ in the mild formula and iterating the resulting singular Gronwall inequality gives a uniform Hessian bound; the equation then bounds V_t^α . More explicitly, with $M(r) = \sup_{s \leq r} \|D^2 W(s)\|_\infty$, the differentiated mild equation yields $M(r) \leq C + C \int_0^r (r - s)^{-1/2} (1 + M(s)) ds$. Iterating once converts the weakly singular kernel into an ordinary Gronwall inequality, giving $\sup_{r \leq T} M(r) \leq C_T$ with constants independent of α and $\nu_G \in \mathcal{V}_{[0, T]}$. Related affine-control estimates appear in [8]. Every bounded classical solution satisfies the same mild formula, so uniqueness also follows from the contraction construction (equivalently from [10,

Ch. I, Sec. 2, Thm. 2.6]). $\blacksquare$

Corollary 5 (Closed-Loop Well-Posedness): For any fixed $\nu_G \in \mathcal{V}_{[0, T]}$, the optimal feedback law is given by,

$$\phi_\alpha(t, x | \nu_G) := \arg \min_{u \in U_\alpha(t, \bar{\lambda}_\alpha(t))} \left\{ \tilde{f}(x, u, \nu_G) \cdot \nabla V^\alpha(t, x) + \tilde{l}(x, u, \nu_G) \right\} \quad (29)$$

is uniformly bounded and globally Lipschitz continuous with respect to x . Consequently, the following closed-loop SDE,

$$dx_\alpha(t) = \tilde{f}(x_\alpha(t), \phi_\alpha(t, x_\alpha(t) | \nu_G), \nu_G(t); g_\alpha) dt + \sigma dw_\alpha(t) \quad (30)$$

satisfies the standard Itô conditions and admits a unique strong solution.

Proof: The uniform boundedness of ϕ_α is guaranteed by the compactness of the constraint set K (H2). By (H1), the argmin is a singleton. Since $\tilde{f}$ and $\tilde{l}$ are Lipschitz in x and u , and ∇V^α is Lipschitz in x by Lemma 4. Let $P = C_V$, where C_V is the uniform bound on ∇V^α from Lemma 4. By (H1),

$$\begin{aligned} & |\phi_\alpha(t, x | \nu_G) - \phi_\alpha(t, y | \nu_G)| \\ & \leq L_\alpha(P) (|x - y| + |\nabla V^\alpha(t, x) - \nabla V^\alpha(t, y)|) \\ & \leq L_\alpha(P) (1 + C_V) |x - y|. \end{aligned} \quad (31)$$

Thus the feedback is globally Lipschitz in x , uniformly in α and the admissible flow ν_G . Thus the Lipschitz property of the drift ensures a unique strong solution to the associated SDE. $\blacksquare$

IV. EXISTENCE OF THE CONSISTENT JOINT MEASURE FLOW

In this section, we establish the existence of a fixed point in the space of joint probability measure flows. The primary challenge in the GMFGC framework is that the control constraints depend dynamically on the network control mean field, $\bar{\lambda}_\alpha(t)$, which in turn is governed by the joint measure flow ν_G .

We begin by defining the appropriate metric space for the adjoint measure flow and the pushforward mechanism induced by the optimal feedback laws. Recall that $\mathcal{V}_{[0, T]}$ is equipped with the uniform Wasserstein metric d_V in (14).

Definition 6 (Joint Measure Disintegration): For any fixed $\nu_G \in \mathcal{V}_{[0, T]}$, let $\phi_\alpha(t, x | \nu_G)$ be the unique optimal feedback law derived from (29). If $\mu_{\alpha, t}$ is the marginal state measure, the consistent joint measure $\nu_{\alpha, t}^{\text{new}}$ is defined by the disintegration,

$$\begin{aligned} & \int_{\mathbb{R}^n \times \mathbb{R}^m} \psi(x, u) \nu_{\alpha, t}^{\text{new}}(dx, du) \\ & = \int_{\mathbb{R}^n} \psi(x, \phi_\alpha(t, x | \nu_G)) \mu_{\alpha, t}(dx) \end{aligned} \quad (32)$$

for any bounded continuous test function ψ . Equivalently, $\nu_{\alpha, t}^{\text{new}} = (\text{Id}, \phi_\alpha(t, \cdot | \nu_G))_\# \mu_{\alpha, t}$. $\square$

To establish a contraction mapping we employ the following uniform sensitivity condition; it is the constrained graphon

analogue of the best-response stability hypothesis used in standard MFG contraction arguments.

(H8) [Uniform Best-Response Sensitivity] There exists a constant $C_\phi > 0$ such that, for every $\nu_G, \nu'_G \in \mathcal{V}_{[0,T]}$, the feedback laws generated by their respective HJB equations satisfy

$$\sup_{\alpha \in [0,1]} \sup_{t \in [0,T], x \in \mathbb{R}^n} |\phi_\alpha(t, x | \nu_G) - \phi_\alpha(t, x | \nu'_G)| \leq C_\phi d_V(\nu_G, \nu'_G). \quad (33)$$

For each fixed admissible flow, the feedback is also uniformly η -Hölder in time and continuous in α on bounded state sets.

Remark 7: Assumption (H8) controls the complete best response, including the moving feasible set, flow-dependent Hamiltonian, and HJB costate. It therefore does not follow from Hausdorff stability (H3) alone. In the finite-dimensional moment setting of [8], (33) follows from uniform strong regularity of the parameterized KKT system, under uniform LICQ and second-order sufficiency, together with Lipschitz stability of the HJB costate. Since (H1)–(H7) do not ensure these properties, we impose (H8) separately.

We next construct the two-level fixed-point argument. We first define the marginal projection from path laws to joint measure flows and then the inner map on the path space. Given a probability measure $\mathbf{P}_\alpha \in \mathbf{V}_{\alpha,T}$, define the time marginal $\nu_\alpha(t)$ by

$$\nu_\alpha(t, B) := \mathbf{P}_\alpha((x_\alpha(\cdot), u_\alpha(\cdot)) \in C_T : (x_\alpha(t), u_\alpha(t)) \in B),$$

for any Borel set $B \subset \mathbb{R}^{n+m}$. We denote this joint marginal by

$$\text{Marg}_t(\mathbf{P}_\alpha) = \nu_\alpha(t). \quad (34)$$

Its state marginal is $\mu_{\alpha,t}^{\mathbf{P}} = (\text{proj}_x)_\# \text{Marg}_t(\mathbf{P}_\alpha)$. Consider a path measure ensemble $\mathbf{P} \in \mathbf{V}_T^G$. Define the time marginal mapping for the entire ensemble by,

$$\text{Marg}_t(\mathbf{P}) = (\text{Marg}_t(\mathbf{P}_\alpha))_{\alpha \in [0,1]}. \quad (35)$$

We use two fixed-point levels. For a prescribed conjectured flow ν_G , the inner map below updates a trial path-law ensemble while keeping ν_G fixed. Its fixed point enforces the McKean–Vlasov consistency of the closed-loop dynamics. The outer map then enforces consistency between the conjectured and realized environments.

Inner MV map. For any fixed assumed joint measure flow $\nu_G \in \mathcal{V}_{[0,T]}$, let $\phi_\alpha(t, x | \nu_G)$ denote the unique optimal feedback law derived from (29). We define the mapping $\Phi_V^{\nu_G} : \mathbf{V}_T^G \rightarrow \mathbf{V}_T^G$ as follows. For any test path measure ensemble $\mathbf{P}_G \in \mathbf{V}_T^G$, we construct the induced joint measure flow,

$$\bar{\nu}_{\alpha,t} := (\text{Id}, \phi_\alpha(t, \cdot | \nu_G))_\# \mu_{\alpha,t}^{\mathbf{P}}. \quad (36)$$

And solve the following closed-loop SDE for all $\alpha \in [0,1]$,

$$dx_\alpha(t) = \tilde{f}(x_\alpha(t), \phi_\alpha(t, x_\alpha(t) | \nu_G), \bar{\nu}_G[\mathbf{P}_G](t); g_\alpha) dt + \sigma dw_\alpha(t), \quad (37)$$

with $x_\alpha(0) \sim \mu_0$. The output $\mathbf{P}_G^{\text{new}} = \Phi_V^{\nu_G}(\mathbf{P}_G)$ is defined as the path law ensemble of the solutions $(x_\alpha(\cdot), \phi_\alpha(\cdot, x_\alpha(\cdot) | \nu_G))$.

Lemma 8 (Uniqueness of the Consistent Path Element):

Under assumptions (H1)–(H8), for any fixed $\nu_G \in \mathcal{V}_{[0,T]}$, the mapping $\Phi_V^{\nu_G}$ admits a unique fixed point $\mathbf{P}_G^{\nu_G} \in \mathbf{V}_T^G$. We define $\mathbf{V}_T^{G1} \subset \mathbf{V}_T^G$ as the set of all such unique fixed points generated by all $\nu_G \in \mathcal{V}_{[0,T]}$. Furthermore, the marginal flow of any element in $\mathbf{V}_T^{G1}$ belongs to $\mathcal{V}_{[0,T]}$.

Proof: Let $\mathbf{P}_G, \mathbf{Q}_G \in \mathbf{V}_T^G$. For a 1-Lipschitz $\chi(x, u)$, the map $x \mapsto \chi(x, \phi_\alpha(t, x | \nu_G))$ is $(1 + L_\phi)$ -Lipschitz. Hence

$$\sup_\alpha W_1(\bar{\nu}_{\alpha,t}[\mathbf{P}_G], \bar{\nu}_{\alpha,t}[\mathbf{Q}_G]) \leq (1 + L_\phi) D_t(\mathbf{P}_G, \mathbf{Q}_G). \quad (38)$$

Because D_t is a joint state–control path metric, write $P_\alpha^x = (\text{proj}_x)_\# \mathbf{P}_\alpha$ and define

$$D_t^x(\mathbf{P}_G, \mathbf{Q}_G) := \sup_\alpha W_{1, \rho_t^x}(P_\alpha^x, Q_\alpha^x), \quad \rho_t^x(x, x') := \sup_{s \leq t} |x_s - x'_s|. \quad (39)$$

Couple the two output SDEs with the same initial state and Brownian motion. Set $\hat{\mathbf{P}} = \Phi_V^{\nu_G}(\mathbf{P}_G)$ and $\hat{\mathbf{Q}} = \Phi_V^{\nu_G}(\mathbf{Q}_G)$. The Lipschitz bound for $\tilde{f}$ and the preceding estimate give

$$D_t^x(\hat{\mathbf{P}}, \hat{\mathbf{Q}}) \leq L_f(1 + L_\phi) \int_0^t D_s^x(\hat{\mathbf{P}}, \hat{\mathbf{Q}}) ds + L_f(1 + L_\phi) \int_0^t D_s(\mathbf{P}_G, \mathbf{Q}_G) ds. \quad (40)$$

Since both output controls use the same feedback, $D_t(\hat{\mathbf{P}}, \hat{\mathbf{Q}}) \leq (1 + L_\phi) D_t^x(\hat{\mathbf{P}}, \hat{\mathbf{Q}})$. Gronwall's lemma therefore yields

$$D_t(\Phi(\mathbf{P}_G), \Phi(\mathbf{Q}_G)) \leq A \int_0^t D_s(\mathbf{P}_G, \mathbf{Q}_G) ds, \quad A := L_f(1 + L_\phi)^2 e^{L_f(1 + L_\phi)T}. \quad (41)$$

Iterating gives $D_T(\Phi^k(\mathbf{P}_G), \Phi^k(\mathbf{Q}_G)) \leq (AT)^k D_T(\mathbf{P}_G, \mathbf{Q}_G)/k!$. Thus some iterate is a strict contraction, and Φ has a unique fixed point.

Bounded drift and constant diffusion give $1/2$ -Hölder state marginals in W_1 . The spatial Lipschitz and temporal Hölder bounds in (H8) give $\mathbb{E}|u_\alpha(t) - u_\alpha(s)| \leq C(|t - s|^{1/2} + |t - s|^\eta)$. The resulting universal estimates imply (C2)–(C3), and the common initial law together with (H3), (H6), and the α -continuity in (H8) gives (C1). Choosing the fixed class bounds C_h, M_V no smaller than these estimates makes the inner solution map a self-map of $\mathcal{V}_{[0,T]}$. ■

Accordingly, define the inner solution map

$$\mathcal{S} : \mathcal{V}_{[0,T]} \longrightarrow \mathbf{V}_T^{G1}, \quad \mathcal{S}(\nu_G) := \mathbf{P}_G^{\nu_G}. \quad (42)$$

We now evaluate the sensitivity of these fixed points in $\mathbf{V}_T^{G1}$ with respect to the original assumed flows.

Lemma 9 (Sensitivity of Consistent Path Measures):

Let $\nu_G, \nu'_G \in \mathcal{V}_{[0,T]}$ be two assumed flows, generating best responses $\phi_\alpha, \phi'_\alpha$. Let $\mathbf{P}_G, \mathbf{P}'_G \in \mathbf{V}_T^{G1}$ be their corresponding unique fixed-point path ensembles from

Lemma 8. There exists a constant C_V such that,

$$D_T(\mathbf{P}_G, \mathbf{P}'_G) \leq C_V \sup_{\alpha, t, x} |\phi_\alpha(t, x|\nu_G) - \phi'_\alpha(t, x|\nu'_G)|.$$

Proof: Put $\Delta_\phi = \sup_{\alpha, t, x} |\phi_\alpha(t, x|\nu_G) - \phi'_\alpha(t, x|\nu'_G)|$ and couple the two fixed-point SDEs with the same initial state and Brownian motion. Let $S_t = \sup_\alpha \mathbb{E} \sup_{s \leq t} |x_\alpha(s) - x'_\alpha(s)|$. For their induced joint marginals, the pushforward estimate is

$$\sup_\alpha W_1(\bar{\nu}_{\alpha, t}, \bar{\nu}'_{\alpha, t}) \leq (1 + L_\phi)S_t + \Delta_\phi. \quad (43)$$

The SDE difference and Lipschitz continuity of $\tilde{f}$ then give

$$S_t \leq 2L_f \int_0^t ((1 + L_\phi)S_s + \Delta_\phi) ds,$$

and hence $S_T \leq B\Delta_\phi$, where $B := 2L_f T e^{2L_f(1+L_\phi)T}$. The path metric D_T also contains the control coordinate; using $|\phi_\alpha(t, x) - \phi'_\alpha(t, y)| \leq L_\phi|x - y| + \Delta_\phi$ gives

$$\begin{aligned} D_T(\mathbf{P}_G, \mathbf{P}'_G) &\leq (1 + L_\phi)S_T + \Delta_\phi \\ &\leq C_V \Delta_\phi, \quad C_V := 1 + (1 + L_\phi)B. \end{aligned} \quad (44)$$

We now define the contraction mapping on the joint measure flow space.

Definition 10 (Outer Consistency Map): Set

$$\mathbf{K}_T^G := \{\mathbf{P}_G \in \mathbf{V}_T^G : \text{Marg}(\mathbf{P}_G) \in \mathcal{V}_{[0, T]}\}. \quad (45)$$

Define $\Psi : \mathbf{K}_T^G \rightarrow \mathbf{K}_T^G$ by

$$\Psi := \mathcal{S} \circ \text{Marg}. \quad (46)$$

Equivalently, for every $\mathbf{P}_G \in \mathbf{K}_T^G$,

$$\Psi(\mathbf{P}_G) = \mathcal{S}(\text{Marg}(\mathbf{P}_G)). \quad (47)$$

Thus the inner map holds the conjectured flow fixed, whereas the outer map updates that flow using the realized joint marginals.

(H9) The model parameters satisfy the weak-coupling condition $C_\phi C_V < 1$, where C_ϕ and C_V are the constants in (H8) and Lemma 9, respectively.

Theorem 11 (Generalized Nash equilibrium): Under assumptions (H1)–(H9), the mapping Ψ admits a unique fixed point $\mathbf{P}_G^* \in \mathbf{K}_T^G$. The induced flow $\nu_G^* = \text{Marg}(\mathbf{P}_G^*)$, together with the feedback strategy $\phi_\alpha(t, x|\nu_G^*)$, constitutes a unique generalized Nash equilibrium for the GMFGC within the admissible Markov-feedback class.

Proof: The marginal projection is 1-Lipschitz from path-law distance to the uniform W_1 flow distance. Hence the defining conditions (C1)–(C3), with fixed bounds, are closed under D_T convergence, so $\mathbf{K}_T^G$ is complete. Constant-path ensembles show that it is nonempty. Let $\mathbf{P}_G, \tilde{\mathbf{P}}_G \in \mathbf{K}_T^G$, and denote their corresponding joint marginal flows by $\nu_G = \text{Marg}(\mathbf{P}_G)$ and $\tilde{\nu}_G = \text{Marg}(\tilde{\mathbf{P}}_G)$. By Lemma 9, the distance between the mapped path ensembles is bounded by

the difference in their induced feedback policies,

$$\begin{aligned} D_T(\Psi(\mathbf{P}_G), \Psi(\tilde{\mathbf{P}}_G)) &\leq C_V \sup_{\alpha, t, x} |\phi_\alpha(t, x|\nu_G) - \phi_\alpha(t, x|\tilde{\nu}_G)|. \end{aligned} \quad (48)$$

By the uniform best-response sensitivity assumption (H8),

$$D_T(\Psi(\mathbf{P}_G), \Psi(\tilde{\mathbf{P}}_G)) \leq C_\phi C_V d_V(\nu_G, \tilde{\nu}_G). \quad (49)$$

The time-marginal map is 1-Lipschitz from the joint path metric to the joint state-control W_1 metric, which implies $d_V(\nu_G, \tilde{\nu}_G) \leq D_T(\mathbf{P}_G, \tilde{\mathbf{P}}_G)$. Substituting this into (49) yields,

$$D_T(\Psi(\mathbf{P}_G), \Psi(\tilde{\mathbf{P}}_G)) \leq C_\phi C_V D_T(\mathbf{P}_G, \tilde{\mathbf{P}}_G). \quad (50)$$

Under assumption (H9), $C_\phi C_V < 1$. Therefore, Ψ is a strict contraction on the complete metric space $\mathbf{K}_T^G$. By the Banach Fixed-Point theorem, there exists a unique fixed point $\mathbf{P}_G^* = \Psi(\mathbf{P}_G^*)$. This fixed point guarantees consistency between the assumed environment and the realized state-control distribution. At that environment, Lemma 4 and its verification argument show that ϕ_α is feasible and optimal for every node. Thus the fixed point is the claimed generalized Nash equilibrium; uniqueness holds within the stated admissible Markov-feedback class. ■

V. A NUMERICAL EXAMPLE

We use the upgraded smart-grid HVAC test in which x is temperature deviation, u is HVAC power, and θ_α is the graphon-weighted neighborhood mean control. Here $\theta_\alpha(t) = \int_0^1 \mathcal{W}(\alpha, \beta) \int u \nu_{\beta, t}(dx, du) d\beta$. The model is

$$\begin{aligned} dx_\alpha(t) &= [-a \tanh^2(x_\alpha(t)) + bu_\alpha(t) + c\theta_\alpha(t)] dt \\ &\quad + \sigma dW_\alpha(t), \end{aligned}$$

$$U_\alpha(t) = [\theta_\alpha(t) - \Delta, \theta_\alpha(t) + \Delta] \cap [u_{\min}, u_{\max}],$$

The running cost is $q(x) + r_1 u^2 + r_2 (u - \theta_\alpha)^2$ and the terminal cost is $q_T(x)$, where the bounded smooth comfort penalties q and q_T are centered at -1.5 and 1.5 , respectively. We take $\mathcal{W}(\alpha, \beta) = 2\alpha\beta$, $a = .3$, $b = 1$, $c = .6$, $\sigma = .45$, $[u_{\min}, u_{\max}] = [-.5, .5]$, $\Delta = .4$, $r_1 = r_2 = .02$, and $T = 1$. To specify the bounded comfort terms, set $Q_{z, M}(x) := M \tanh(5(\sqrt{(x - z)^2 + .1^2} - .1)/M)$; we use $q = Q_{-1.5, 80}$ and $q_T = Q_{1.5, 15}$. Thus the state costs meet (H4), while the quadratic control and conformity terms are bounded and Lipschitz on the compact control and mean-field ranges; equivalently, they may be continued outside those ranges by bounded Lipschitz extensions. The term $-a \tanh^2(x)$ is intentional: it is a bounded downward thermal-load drift, rather than an odd mean-reverting drift. For this scalar example the interval endpoints $\max\{\theta - \Delta, u_{\min}\}$ and $\min\{\theta + \Delta, u_{\max}\}$ are Lipschitz in θ , so (H3) holds. Since $|u| \leq .5$ and $\int_0^1 \mathcal{W}(\alpha, \beta) d\beta = \alpha \leq 1$, one has $\theta_\alpha \in [-.5, .5]$ and the feasible interval is nonempty. The quadratic control terms make the pointwise Hamiltonian strongly convex with modulus $2(r_1 + r_2)$. With $p = V_x^\alpha(t, x)$, $\ell_\theta = \max\{\theta - \Delta, u_{\min}\}$, and $h_\theta = \min\{\theta + \Delta, u_{\max}\}$, its

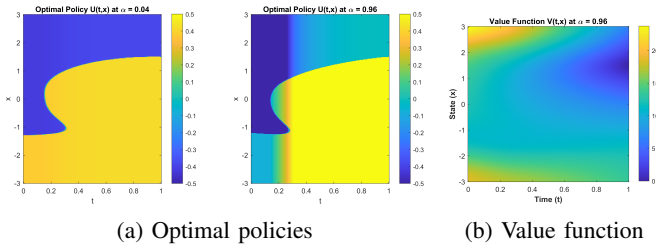


Fig. 1: Optimal policies and value function under the moving control constraint.

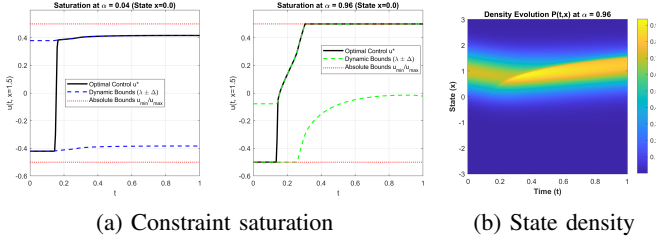


Fig. 2: Constraint activity and state-density evolution for representative nodes.

optimizer is

$$u^0(p, \theta) = \frac{2r_2\theta - bp}{2(r_1 + r_2)}, \quad \phi_\alpha(t, x) = \Pi_{[\ell_\theta, h_\theta]} u^0(p, \theta). \quad (51)$$

The endpoint maps and scalar projection are Lipschitz; hence (51) verifies the uniqueness and pointwise Lipschitz requirement in (H1). The HJB–FPK iteration uses $N_x = 301$, $N_t = 201$, $N_\alpha = 12$, and 60 mean-field updates on $x \in [-3, 3]$; the common initial density is the Gaussian $\mathcal{N}(1, .55^2)$, truncated and normalized on this interval. We impose homogeneous Neumann conditions for the HJB and a reflecting zero-flux condition for the FPK equation. The diffusion step is implicit, advection is upwinded, and the FPK flux is conservative. Starting from $\theta^{(0)} = 0$, each mean-field update is relaxed according to $\theta^{(k+1)} = .2 \Theta(\theta^{(k)}) + .8 \theta^{(k)}$, where Θ denotes one HJB–FPK and graphon update. We run exactly 60 updates and monitor the unrelaxed residual $R_k = \sup_{\alpha, t} |\Theta(\theta^{(k)})_\alpha(t) - \theta_\alpha^{(k)}(t)|$; no early stopping is used. The four panels in Figs. 1–2 show the active moving constraint, its effect on the population, and the resulting value and density fields.

For iteration k , define $e_k = \sup_{\alpha, t} |\theta_\alpha^{(k+1)}(t) - \theta_\alpha^{(k)}(t)|$. It decreases monotonically from about 4×10^{-1} to below 10^{-3} over the 60 updates. This is a diagnostic of the discretized scheme, not a substitute for the analytical weak-coupling condition. The conservative FPK flux preserves discrete mass up to solver tolerance, and projection onto the moving interval enforces pointwise feasibility. These checks establish stabilization and feasibility only for the stated discretization. In particular, neither the decay of e_k nor the discrete mass balance is used to infer (H8) or the analytical inequality in (H9), whose constants must be uniform before discretization. The computation is therefore an illustration of the equilib-

rium mechanism, while the theorem rests exclusively on the continuous assumptions and estimates above.

VI. CONCLUSION AND FUTURE WORK

This work establishes existence and uniqueness, within the admissible Markov-feedback class, of a consistent joint measure flow for a GMFGC with moving control constraints. Hausdorff stability provides minimized-Hamiltonian regularity, whereas full-environment optimizer stability is imposed separately in (H8), avoiding the inference of optimizer Lipschitz continuity from set convergence alone. Two limitations remain. First, uniqueness holds only in this feedback class under (H9), not among open-loop or relaxed equilibria. Second, the numerical example is illustrative, and convergence of the HJB–FPK discretization remains for future work. Deriving (H8)–(H9) from primitive data beyond [8] remains an important direction.

Future work should address failures of Hamiltonian regularity through viscosity or weak solutions. Direct distributional constraints would model density limits and congestion penalties, while sparse graphexon limits would represent heterogeneous network sparsity more faithfully [12], [13]. Finally, the ϵ -Nash property is needed to validate these strategies in finite-agent systems of size N .

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